### *Setting a default catalog and project-specific database*

This book provides code that uses the Unity Catalog catalog. Your default catalog is set based on your environment. If you do not set the environment or you set it to `dev`, then the catalog is named `ml_in_action`. When the environment is `prod`, the catalog is `ml_in_prod`. The default name for the database is always the project name. However, you can provide a different name if you desire by entering a project variable for the database name.

```
for i in range(10):
  try:
    spark.sql(f"""USE `{catalog}`.`{database_name}`""")
    break
  except Exception as e:
    time.sleep(1)
    if i >= 9:
      raise e
```

Figure 2.16 – Using the defined variables to set the default with retries

We want to be sure that the catalog and database are set to the notebooks' defaults. Occasionally, with parallel execution, this command can fail during initialization; therefore, we add a few retries to work around this issue, as shown in *Figure 2.16*.

### *Granting permissions*

Now that we've set our catalog and database defaults, we can grant permissions.

```
# Granting UC permissions to account users - change if you want your data private
spark.sql(f"GRANT CREATE, SELECT, USAGE on SCHEMA {catalog}.{database_name} TO `account users`")
```

Figure 2.17 – Granting permissions to the catalog and database

We grant the group `account users` permission. If you do not want to make your assets available to others, remove this or comment it out.

> **Important note**
>
> Be sure to use tick marks around your group name or email address when granting permission. If you use single quotes instead, you will get an error message.

The catalog and database are ready for tables. However, not all data we use in machine learning goes into a table. For other data, files, and objects, we have volumes.